Financial Management (MBACCZG521)

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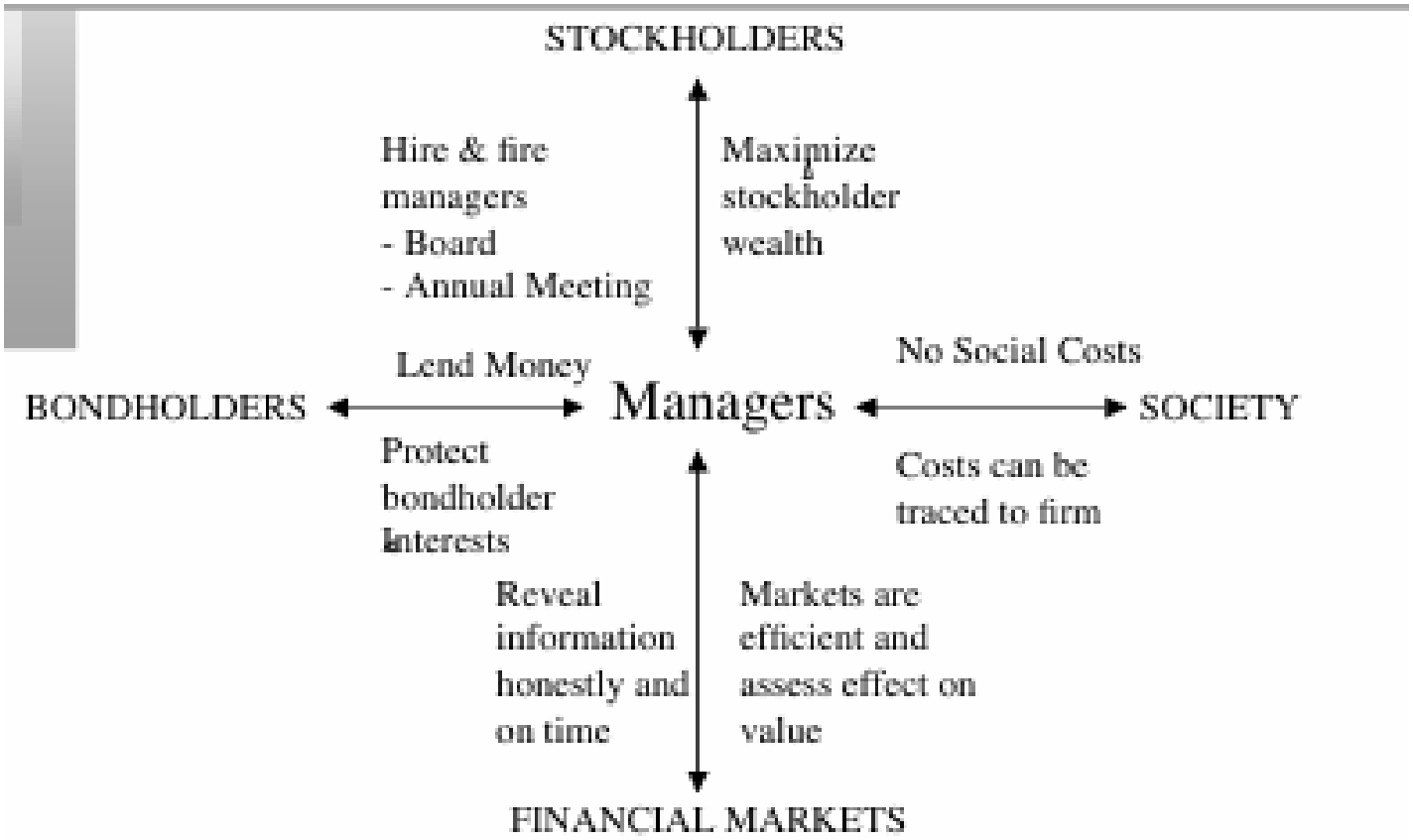
Financial Planning

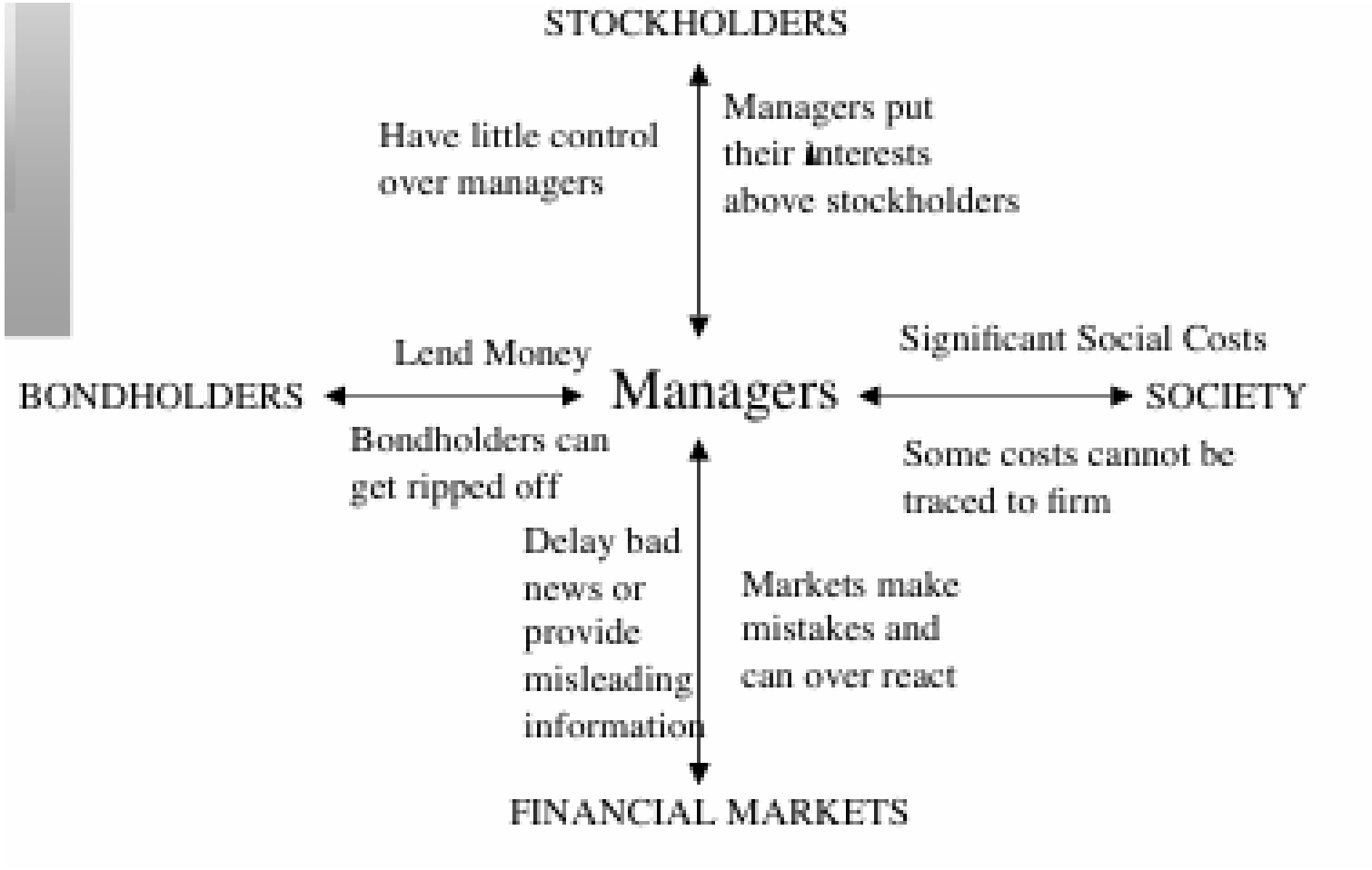
Agency Problem

- Agency relationship
  - Principal hires an agent to represent its interests
  - Stockholders (principals) hire managers (agents) to run the company
- Agency problem
  - Conflict of interest between principal and agent
  - Example: Not pursuing risky project for fear of losing jobs, expensive perks, expanding the business for increasing scope/ scale of power
- All else equal, agency problems will reduce the firm value.

The Classic Objective Function



What can go wrong?



How to Reduce Agency Problems?

1. **Monitoring**

   (Examples: Reports, Meetings, Auditors, board of directors, financial markets, bankers, credit agencies)

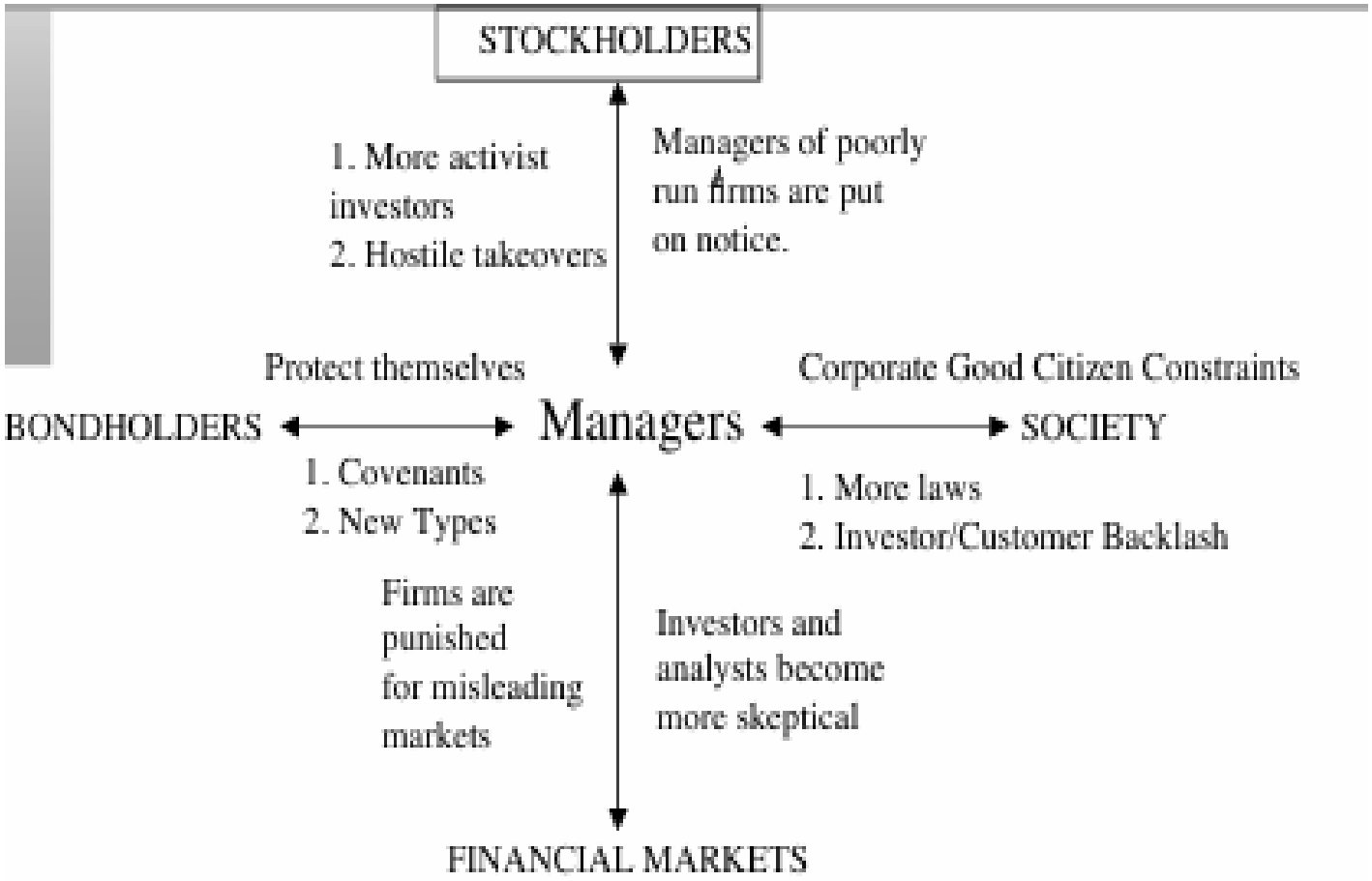
2. **Compensation plans**

   (Examples: Performance based bonus, salary, stock options, benefits)

3. **Others**

   (Examples: Threat of being fired, Threat of takeovers, Stock market, regulations such as SOX)

The Counter Reaction



Ethics in Finance

- What do we mean by Ethics?
- Examples of recent financial scandals
  - Satyam Computers scam
  - JP Morgan Scam
  - Enron scam, etc.

Major Financial Scandals in India

1. **Harshad Mehta Scam (1992) -** ₹5,000 crore
2. **Satyam Scam (2009) -**
3. **Ketan Parekh Scam (2001)**
4. **PNB Nirav Modi Scam (2018)**
5. **Sahara Scam (2010) -** ₹24,000 crore
6. **Saradha Chit Fund Scam (2013)**
7. **Vijay Mallya and Kingfisher Airlines Scam (2016) -** ₹9,000 crore
8. **PMC Bank Scam (2019) -** ₹6,500
9. **SpeakAsia Scam (2010) -** 24 lakh investors
10. **Adarsh Housing Society Scam (2010)**

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Financial forecasting and long-term planning – Time Series analysis

Time Series Analysis

A *time-series* is a set of observations on a quantitative variable collected over time.

Examples

- Dow Jones Industrial Averages
- Historical data on sales, inventory, customer counts, interest rates, costs, etc

Businesses are often very interested in forecasting time series variables.

Often, independent variables are not available to build a regression model of a time series variable.

In time series analysis, we analyze the past behavior of a variable in order to predict its future behavior.

Methods used in Forecasting

- Regression Analysis
- Time Series Analysis (TSA)
  - A statistical technique that uses time-series data for explaining the past or forecasting future events.
  - The prediction is a function of time (days, months, years, etc.)
  - No causal variable; examine past behavior of a variable and attempt to predict future behavior

Methods of Time-Series Analysis

Four key components of time series:

- Secular or long term trend
- Cyclical – significant fluctuations around the long term trend over an extended period of time (over a year)
- Seasonality – variations due to seasonality occur at about the same time every year (Christmas season)
- Irregular variations – fluctuations that do not belong to the above groups
  - Variations due to random events or episodes (such as opening of Soviet Union)
  - Residual variations or noise (not associated with any of the above)

Measuring linear secular trends

- Linear regression analysis: y(t) = a + bt
- Y(t) is the value of the time series at time = t
- A is the intercept or the value of Y(t) at time t = 0
- B is the regression co-efficient or the slope of the line

Using least squares method we can determine these parameters

- B = [(n*sigma(tY)) – (sigmaY)*sigma(t)]/[(n*sigma(t^2))-(sigma(t)^2)]
- A = Ybar – b*tbar

Moving average

- Applicable to time series where the secular trend is linear and the variations are periodic
- Moving average is a way to smoothen the cyclical and seasonal variations
- Cetered vs. uncentered moving averages
- It is a good proxy for the long term secular trend

Centered moving average means it is centered for the period, for example if it is a 3 year moving average the first observation would be the average of the first three years and it would be centered at year 2; if it is non-centered then it would start at year 3

Exercise

Seasonality

THANK YOU!!!